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Wideband circularly polarized dielectric resonator antenna with high gain for microwave wireless power transfer

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Wireless power transmission (WPT) is poised to revolutionize the future of wireless applications and sensing networks. High-gain antennas are essential for extending WPT coverage, but the unpredictable orientation of wireless devices remains a major challenge. To address this, this paper proposes a dielectric resonator antenna (DRA) with a new cone version (cupped-cone shape) to achieve high gain and a wideband circular polarization, ensuring consistent energy transfer regardless of device orientation. The proposed design enhances bandwidth by supporting multiple resonant modes. Combining two geometric shapes provides greater flexibility for fine-tuning and optimizing the DRA. The proposed DRA is excited using an innovative feeding mechanism with two elliptical slots and a modified microstrip feeding to produce wideband circular polarization, achieving large impedance and axial ratio bandwidths. The design is fabricated from polylactic acid using 3D printing technology, making it lightweight and cost-effective. Measurements show the antenna operates in the WPT band, covering the industrial, scientific, and medical (ISM) frequency of 5.8 GHz with a 64% impedance bandwidth, 3-dB axial ratio bandwidth of approximately 31%, and a high gain of about 11.1 dBic by effectively utilizing a higher-order mode while maintaining the bandwidth.

The rapid evolution of wireless communication technology has created a growing need for innovative wireless devices. Traditional wired power supplies are no longer suitable, as they restrict mobility and are inefficient for powering devices in hard-to-reach locations. In this context, wireless power transmission (WPT) may beat the challenges of wired power supplies and offer strong potential for applications, such as solar power satellites, the internet of things (IoT), and wireless sensor networks (WSNs)[1].

Nikola Tesla's work more than a century ago laid the foundation for WPT using electromagnetic (EM) radiation. His studies on microwave transmission and inductive coupling formed the basis of modern WPT systems[2,3]. In general, EM-based WPT can be classified into near-field and far-field techniques, depending on the coupling mechanism. Antennas are a key component of the far-field WPT systems. At the transmitter, the antenna converts electrical power into electromagnetic waves that propagate over long distances. At the receiver, another antenna captures the radiation, which is then rectified and converted into DC power. This conversion is achieved using an efficient rectifier circuit to ensure a high radio-frequency-to-direct-current (RF–DC) conversion efficiency.

To achieve high overall transmission efficiency, antennas used in WPT systems require high radiation gain (hereafter, gain) to effectively direct and transfer the maximum amount of energy to the receiver. Other important system requirements include circular polarization, wide bandwidth, compact size, and flexibility[4].

Circularly polarized (CP) antennas are preferred in WPT because they improve power transfer efficiency. In contrast, misalignment between the transmitter and receiver in linearly polarized (LP) antennas reduces coverage range and efficiency. LP antennas also cause multipath interference, since reflected signals can arrive with different polarizations or from different angles. It should be noted that environmental conditions or mobility can also cause attenuation, which directly reduces the efficiency of LP antennas. The CP antennas, on the other hand, minimize alignment requirements between the transmitting and receiving antennas. This advantage is

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especially important in applications involving device mobility, such as smartphones, drones, satellites, and medical IoT devices.

Microstrip patch antennas have been widely investigated for use in WPT systems because of their compact size and ease of fabrication[5]. However, their relatively low gain and narrow bandwidth limit their suitability for high-efficiency WPT applications. In contrast, dielectric resonator antenna (DRAs)can achieve higher gain and wider bandwidth, making them a more suitable choice for such systems[6,7], particularly due to their high radiation efficiency compared to metallic radiators[8]. In addition, DRAs offer several attractive advantages over other antenna types, such as compact size, lightweight, the ability to employ various feeding techniques[9], and easy integration with microwave integrated circuit components[10]. Moreover, DRAs offer a wide range of available materials, low dielectric loss, simple fabrication processes, high design flexibility over a wide frequency range, and the capability to excite different resonant modes[11].

Another appealing feature of DRAs is their ability to operate with LP, CP, or both by adjusting the feeding techniques. This flexibility has a strong impact on bandwidth, radiation pattern, and overall efficiency[12]. For linearly polarized DRAs, several excitation techniques have been reported using different configurations. These include a microstrip feed line[13], a coaxial cable[14], a tapered dielectric rod waveguide[15], and aperture coupling[16]. Among these methods, stripline feed provides a compact structure and effectively isolates the feeding network from the radiating element[17].

Our previous work in[17] achieved a clear improvement in gain and bandwidth while preserving compact size by modifying the DRA shape. However, that design relied on a high relative permittivity ε_r, which increases cost. Another study highlighted the critical role of the feeding technique and its strong influence on DRA performance and bandwidth[18]. Yet, this improvement is beneficial for applications requiring linear polarization, which may not suit many modern devices. Spiral antennas have also been explored to achieve circular polarization. The metallic spiral offers ultra-wideband performance, but it introduces metal losses and significantly increases cost, reaching several thousand euros[19].

Designing a CP DRA requires careful consideration of several factors, including geometry selection, feeding mechanism, excitation mode, and material choice, to ensure the antenna meets required specifications for polarization and efficiency. Despite these improvements, several challenges hinder the DRA's optimal performance, such as limited bandwidth or low gain, reduced efficiency due to energy loss, complex design, or high fabrication cost.

In CP DRA, narrow bandwidth remains a major challenge. A study in[20] investigated generating CP fields for millimeter-wave applications. The design uses two rectangular slots of different lengths forming a cross-slot at the bottom of the DRA for feeding. This structure is complex, requiring multiple layers with both metallic and non-metallic vias. In another work, a rectangular DRA was excited using an SMA probe for far-field WPT and microwave energy harvesting[22]. The study considered both a single DRA and a 5×5 array. This design achieved a narrow bandwidth of approximately 13%, as shown in the reflection coefficient figures. In[22], a dual-band CP antenna uses a cross-slot coupled rectangular DRA to excite different modes. Although the system provides a reasonable 3-dB axial ratio (AR) bandwidth, the peak gains for the two CP modes are low. Additionally, the antenna's large aspect ratio and use of high-permittivity material result in high fabrication costs.

In this study, a novel shape and innovative feeding mechanism enable a DRA to operate in either linear or circular polarization, depending on the feed type. The design provides high gain and wide bandwidth, ensures high radiation efficiency, and allows simple excitation using a single feed point. The antenna is fabricated using 3D-printing material, making it possible to produce the design at low cost[23,25].

The remainder of the paper is organized as follows: "Literature review for high-gain antennas" provides a literature review of high-gain antennas. "Proposed antenna configuration" illustrates the design procedure of the proposed DRA. "Simulation analysis and discussion" presents the simulation findings, supported by the experimental results in "Experimental validation". Finally, the conclusion and final thoughts are drawn in "Conclusion".

Literature review for high-gain antennas

The transmission of electromagnetic radiation in a free space complies with the basic Friis formula described in (1)[26]:

$$P_r = \left(\frac{\lambda}{4\pi D}\right)^2 P_t G_r G_t \tag{1}$$

where P_r is the receiver power, P_t is the transmitter power, G_r and G_t are the receiver and transmitter gains, respectively. The transmission distance can be represented by D, and λ is the wavelength. According to (1), and since the antenna is a primary element in the WPT system, a high-gain antenna is required to achieve high transmission efficiency. As the antenna gain increases, the receiving power will increase. Therefore, this section studies the previous work that presented a high-gain antenna suitable for use in a WPT system.

For example, in[18], our earlier study developed an innovative feeding technique using a cavity-backed slot, which attained a substantial gain of 11 dBi for a single element functioning at the TE_{115} mode. However, this configuration leads to a limited bandwidth of approximately 14.3%. This is similar to the designs in[26], which achieve a gain of 10.4 dBi with a narrow bandwidth of 16.1% at the TE_{115} mode, and in[27], which propose a new shape DRA with a gain of 11.3 dBi and a restricted bandwidth of 16.6%. Another technique to increase the gain of a DRA was discussed in[29]by integrating additional structures. However, this method is complex and increases the overall size. The same limitation arises when employing multiple-input multiple-output (MIMO) techniques[30]. Numerous studies have achieved high gain using antenna arrays, but their large size limits their suitability for WPT applications[31,32].

Several studies have been applied to DRA to achieve CP, for instance, two annular vias inserted into a simple DRA to get the required CP; however, it achieves a narrow 3 dB axial ratio bandwidth of 5.5%[33]. A single-fed DRA was investigated in[34]. Although this design provides a wide CP bandwidth of 49.5%, it is complex and large in size, as it uses four sequentially rotated metallic plates. Another study on a CP DRA using a cross-slot feed and higher order modes TE_{111} and TE_{113} reports a narrow impedance bandwidth of 8.4%, an axial ratio bandwidth of 2.2%, and a low peak gain of 4.3 dBic at the TE_{113} mode[34]. In[36], a transmit array using a novel elliptical and cylindrical DRA (ECDRA) is proposed. By combining a square substrate with two symmetrical ECDRAs, it converts LP to CP, achieving a gain of 1 dBic with a 13.3% bandwidth and a 3-dB AR of 17%. Lately, a wideband DRA made from high-cost material ECCOSTOCK HIK with $\varepsilon_r = 20$ demonstrated dual circular polarization for x-band radar applications. It was initially tested for LP, and CP was later achieved using a new structure that excites two perpendicular HEM modes. For a single element, this resulted in a 34% impedance bandwidth, a 23% AR bandwidth, and a gain of 7.7 dBi[37]. Although higher-order modes help increase CP DRA gain, impedance and AR bandwidth limitations remain. Additional data and performance comparisons of DRAs and other radiators are provided in[37].

Our research focused on developing a feeding mechanism to enhance CP DRA performance in terms of gain, reflection bandwidth, and AR bandwidth, while keeping cost in mind. In many cases, increasing DRA gain reduces bandwidth, and efforts to widen bandwidth can limit gain. The proposed design uses a new feeding system with two elliptical slots to excite the DRA. This design attains high radiation efficiency and gain, along with broad $|S_{11}|$ and 3 dB AR bandwidths. It effectively excites higher-order modes to increase gain while minimally impacting bandwidth. We simulated this design for various resonator heights and fabricated a prototype with a height of λ to excite the HEM_{115} higher-order mode. The main contribution of this work is the introduction of a novel DRA shape and feeding method that enhances CP without compromising gain or efficiency.

Proposed antenna configuration

The novel 3D cupped-DRA geometry operating in the third mode is shown in Fig. 1a. The proposed geometry is divided into three parts: the radiator part, the feeding part, and the substrate. The radiator is made from polylactic acid (PLA) with a relative permittivity of 2.7 and a low loss tangent of 0.008, which is commonly used in 3D printing due to its favourable properties. The DRA is designed to work in the WPT band at 5.8 GHz. The radiator shown in Fig. 1b consists of a conical structure with an upper radius of R_u = 8 mm, a lower radius of R_l = 11.6 mm, and is placed above a flattened torus made of the same material with a radius of R_t = 11.6 mm. A pair of inclined cylindrical holes with an inclination angle of 35° and a diameter of 0.25 mm is drilled on both sides of the resonator.

The feeding mechanism achieves circular polarization using a novel excitation technique on a substrate of 50 mm × 50 mm. The asymmetrical butterfly-shaped (dumbbell-shaped) slot shown in Fig. 1c controls the surface current distribution in the coupling slot and enables precise adjustment of the axial ratio, resulting in pure circular polarization required for this application. The optimized dimensions of two elliptical slots for minimum axial ratio have radii of R_1 = 10 mm, R_2 = 6.5 mm, with ratios of T_1 = 4 mm and T_2 = 2 mm, respectively. Additionally, two interconnected circular slots are etched in the ground plane to suppress sidelobe in the radiation pattern, with a cavity length between them of Ct = 2 mm.

The feeding mechanism comprises a microstrip line with a feed line width of W_f =4.7 mm (to achieve 50Ω impedance) and a feed line length of L_f = 25 mm, supplemented by additional geometric elements (circular, elliptical, and rectangular) integrated into the microstrip line. This key-shaped structure is designed to achieve impedance matching while maintaining the desired axial ratio bandwidth at the target frequency, as shown in Fig. 1d. Finally, the substrate is made of commercial Rogers material with a relative permittivity of ε_r = 2.2 and a loss tangent of 0.0009, with overall dimensions ($W_g \times L_g$) = 50 mm×50 mm and a thickness of H_s=1.5 mm.

Simulation analysis and discussion

Linearly polarized DRA

First, the antenna was studied for linear polarization, using a rectangular slot notched in the ground and a 50Ω microstrip line to feed the DRA. The structure was then designed and simulated using the High-Frequency Structure Simulator (HFSS), which employs the finite-element method (FEM). The design process began by evaluating the cone-shaped DRA performance, referred to as Ant#1 in Fig. 2, using various materials with different dielectric constants. Since the main goal was to achieve high gain and broad bandwidth at a reasonable cost, PLA was selected for fabrication using a 3D printer due to its low cost and low dielectric constant. The initial simulation results showed limited performance, with a gain of 5 dBi in the first mode and a narrow, discrete bandwidth.

Next, a torus was integrated with the cone, forming Ant#2 in Fig. 2. The results were analysed, focusing on changes in field distribution, gain, and bandwidth caused by combining the cone with the torus. This process was continued through Ant#4, with results compared at each stage to optimize the design.

From the E-field perspective, a change in the strength of the field distribution was observed. Next, the cone position was adjusted up and down relative to the torus. It was found that, depending on the distance between them, combined with adjustments to the torus outer radius and the cone lower radius, the gain increased and a new resonance appeared in the bandwidth compared to the cone alone. This improvement can be attributed to the updated shape, which concentrates previously scattered fields at the corners into the centre, resulting in a gain increase of approximately 2 dBi, as shown in Fig. 3a. From the reflection coefficient perspective, the DRA shape modification is also evident, as shown in Fig. 3b. The final field distribution merges the two narrow modes, producing a continuous and wider bandwidth.

The antenna dimensions were assessed and fine-tuned to operate as a linear-polarized DRA, effectively increasing the gain from 5 dBi to 7 dBi in the first mode HEM_{111} with a height of 0.3 λ, as shown in Fig. 4. The

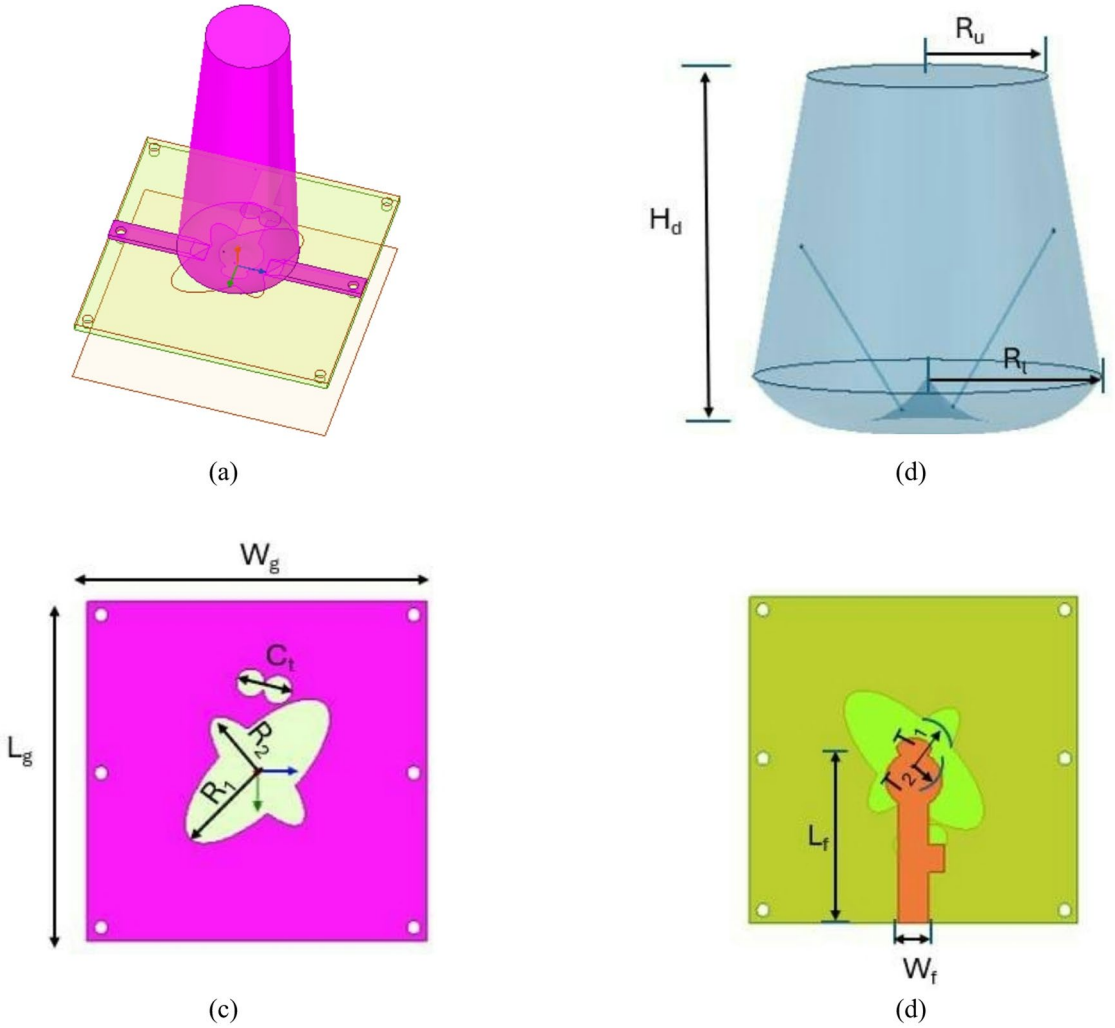


Fig. 1. (**a**) Schematic 3D diagram of the proposed antenna, (**b**) the radiator, (**c**) ground with two elliptical slots, and (**d**) feed line.

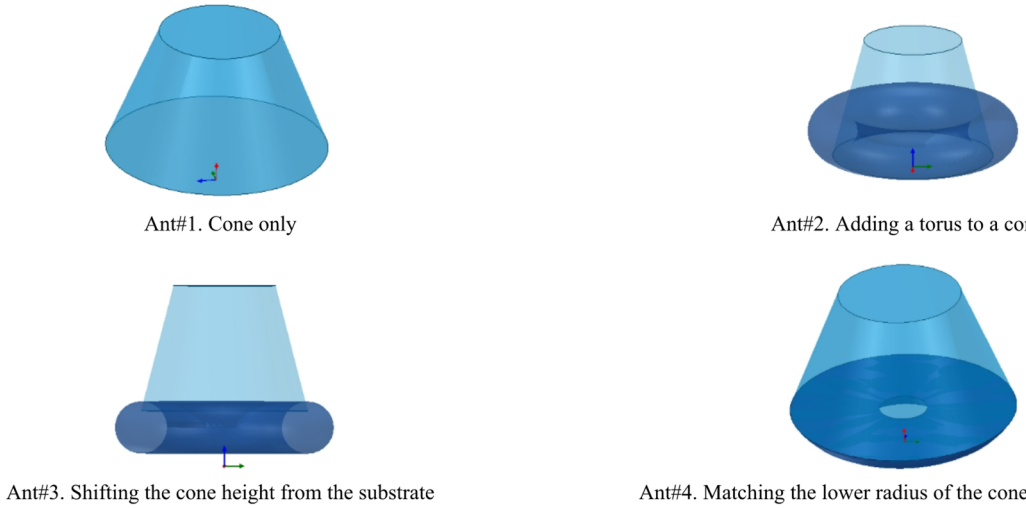


Fig. 2. Design steps of the proposed antenna shape.

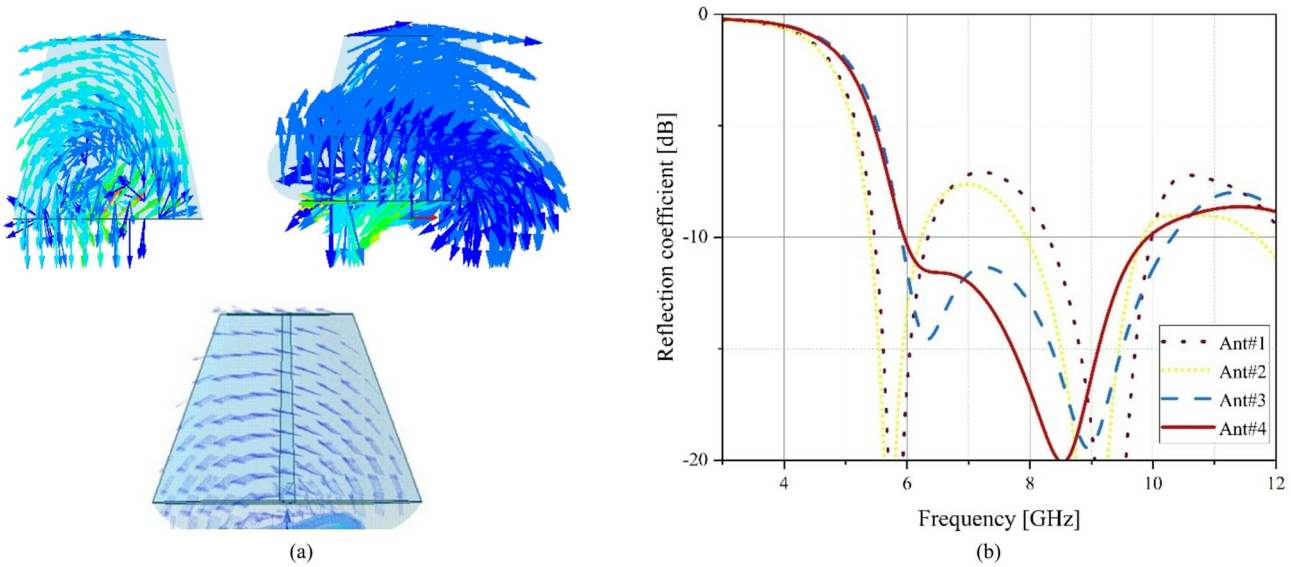


Fig. 3. Impact of various antenna shapes on (**a**) electric field distribution and (**b**) reflection coefficient $|S_{11}|$ versus frequency.

gain can be further increased by raising the DRA height to excite a higher-order mode, achieving about 10 dBi in the HEM_{115} mode along with a broad bandwidth of 60%.

Circularly polarized DRA

To use this DRA in WPT applications, the challenge was to achieve circular polarization while maintaining high gain and a wide bandwidth. Several methods were explored to achieve circular polarization. One approach used conventional rectangular slots, which produced a limited axial ratio of about 12%, considered insufficient, indicating that further improvement was required.

Figure 1c shows a new asymmetric elliptical cross slot rotated by 45°, which has been optimized, as described in the following parametric study section. This optimization ensures operation in the WPT band at 5.8 GHz, enabling circular polarization with a wide AR of approximately 31%. The results, shown in Fig. 5, cover the frequency range from 4.7 to 6.4 GHz. These results demonstrate the effectiveness of our new feeding technique, which outperforms other designs that exhibited lower axial ratio performance compared to our optimized slot shape.

To finalize our design and improve the reflection coefficient, we introduced two cylindrical holes drilled at an inclination angle of 35°, which successfully increased the impedance bandwidth from 44% to approximately 52% through configuration and angle optimization. Additionally, this modification helps tune the axial ratio by adjusting the radii of the two holes. However, our study of the antenna behavior revealed a significant band mismatch between the impedance and axial ratio bandwidths during simulation.

We identified that the dimensions and configuration of the feed line were critical. Therefore, an optimization study was conducted to determine the best length and shape of the microstrip line. This led to the final design shown in Fig. 1d, which achieves a 50 Ω input impedance match. The reflection coefficient $|S_{11}|$ covers the band from 3.3 to 6.4 GHz, as shown in Fig. 6, and aligns well with the AR range from 4.7 to 6.4 GHz, presented earlier in Fig. 5. Both resonances correspond to the TE_{113} mode, as confirmed by the stable radiation patterns, which will be discussed below. Both resonances include the WPT band at 5.8 GHz. The results demonstrate that adjusting the length and shape of the microstrip line effectively tunes the operating frequency to the desired band.

During the simulation phase, another issue was the presence of side lobes and back lobes. To eliminate the side lobes, two connected circular notches were added at a specific location in the ground plane. This modification had a significant impact, completely removing the side lobes, as shown in Fig. 7 with side lobe level (SLL) larger than 14 dB. However, the front-to-back ratio (FBR) remained relatively low at about 8 dB. To improve the radiation pattern, a reflector with the same dimensions as the substrate was introduced. The distance between the reflector and the substrate was optimized to cancel the back lobe, achieving an FBR greater than 15 dB at a distance of 13.44 mm, as shown in Fig. 8. This configuration also slightly increased the gain to 11.1 dBic, as demonstrated in Fig. 9a.

Figure 9b shows the antenna efficiencies of the proposed design, including both the radiation efficiency and the total efficiency (accounting for mismatch losses). The results demonstrate excellent performance with values exceeding 94% across the entire operating bandwidth. This confirms the low-loss nature of the antenna, which is attributed to the absence of lossy elements (such as transitions, phase shifters, metal vias) and the low loss tangent of the radiator material (PLA).

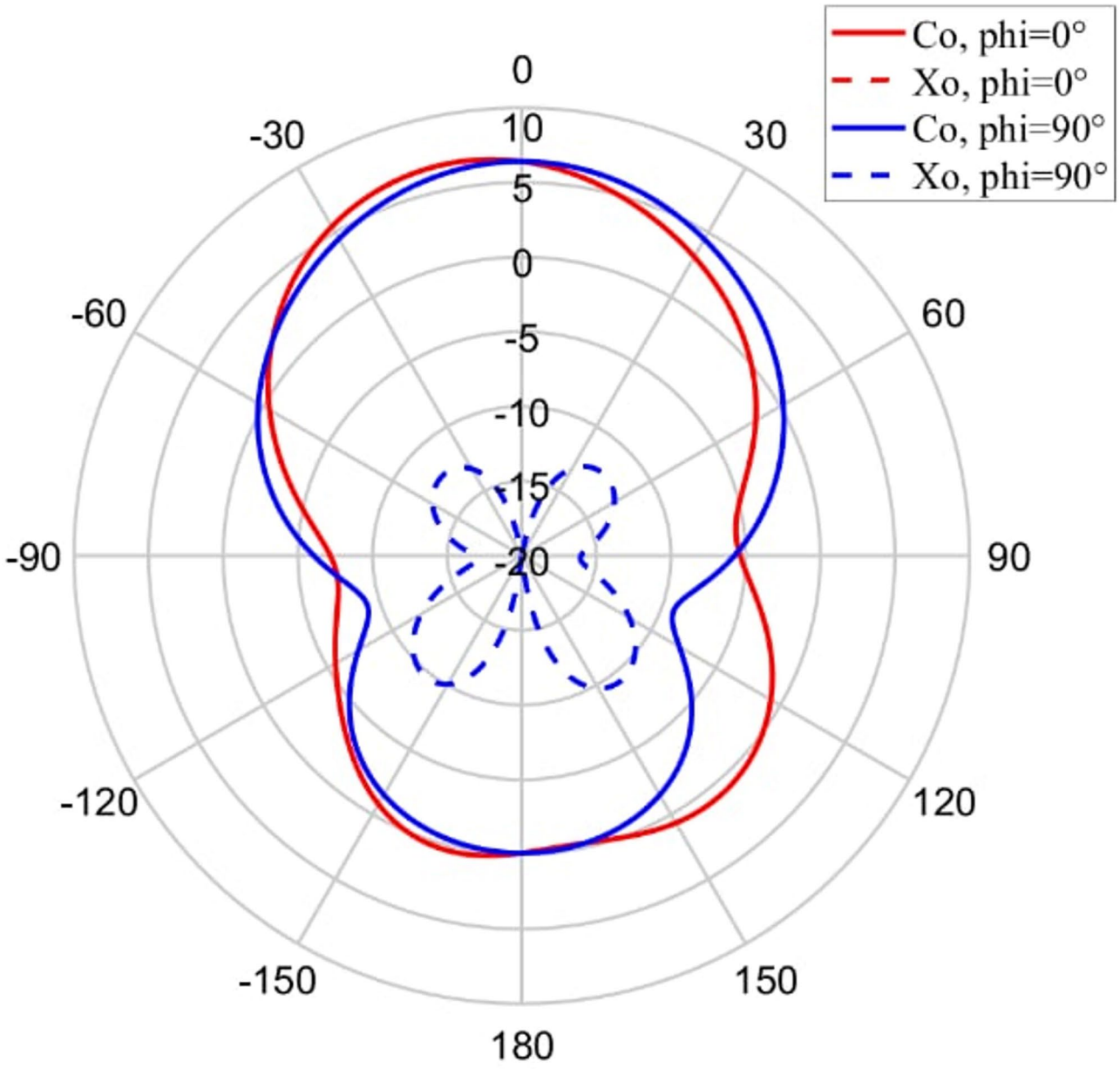


Fig. 4. The simulated far-field radiation pattern of LP DRA.

Design adjustments

In this section, we discuss the main parameters controlling the performance of the proposed design. First, we focus on the CP aspects influenced by the feeding structure, which is the most critical factor affecting the CP performance, particularly the major elliptical slot radius. Figure 10 shows the effect of varying the elliptical slot radius (R_e). The results demonstrate that adjusting R_e allows complete control of the AR shift over frequency, enabling the AR curve to move left or right to tune the desired operating frequency. Another important parameter affecting AR is the outer radius of the torus (R_t). Adjusting R_t shifts the AR curve toward the 3-dB line, optimizing circular polarization. This effect is shown in Fig. 11, where R_t is expressed in millimeters.

Next, the DRA was tested by increasing its height from 0.3 λ to λ. The corresponding plots in Fig. 12 show gain vs. frequency with a peak value of about 7 dBic for the fundamental mode HEM_{111}at a height of 0.3 λ. This is higher than typical basic DRA shapes, which generally achieve around 5 dBi in the same mode. Since WPT applications require a higher gain, exciting higher-order modes was used to achieve this. In our design, the third mode HEM_{115} was excited by increasing the DRA height to λ. The simulation result shows a significant gain increase, with a peak of 10 dBic, which was further improved to 11.1 dBic after final optimization using a reflector. This was achieved while maintaining large and consistent impedance and AR bandwidths. Figure 13 shows the magnetic field (H-field) distribution within the dielectric resonator at 5.8 GHz, explaining the previously observed gain increase. Three DRA heights—18, 33, and 60 mm, corresponding to 0.3λ, 0.6λ, and

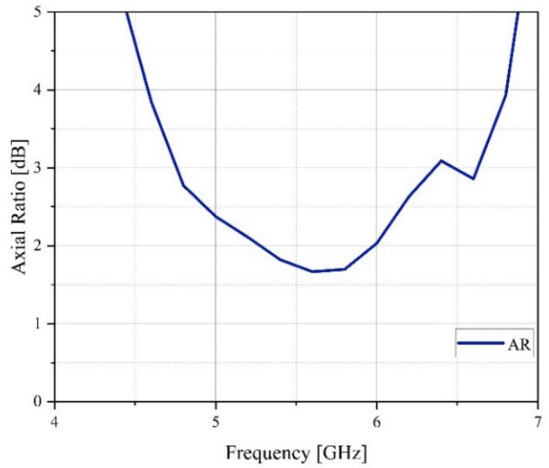


Fig. 5. Axial ratio vs. frequency of CP DRA.

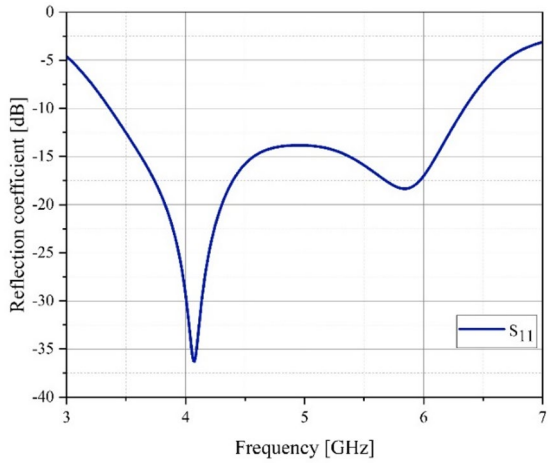


Fig. 6. Reflection coefficient $|S_{11}|$ vs. frequency for CP DRA.

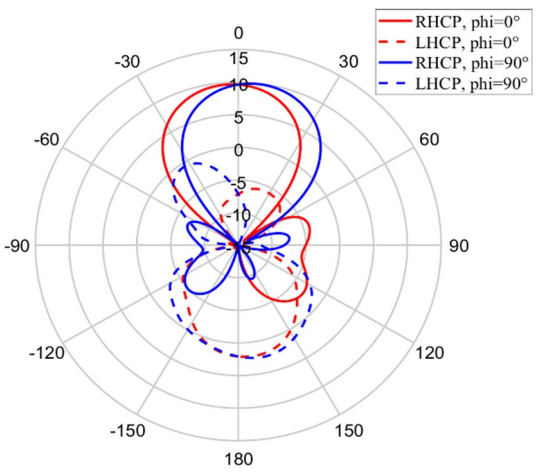


Fig. 7. Simulated radiation pattern at 5.8 GHz.

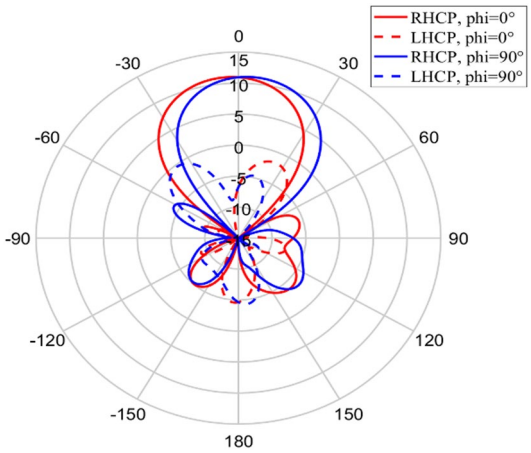


Fig. 8. Simulated radiation patterns at 5.8 GHz with reflector.

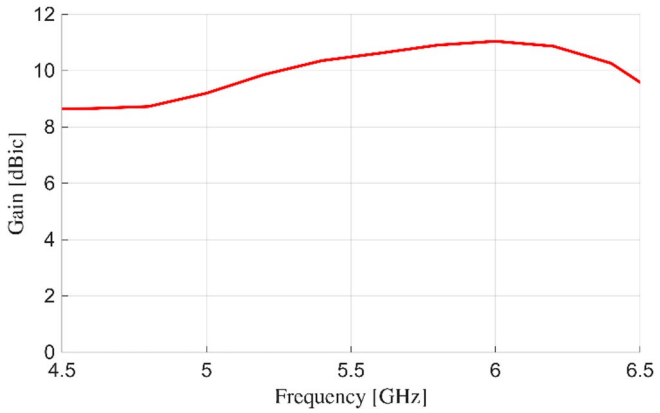


Fig. 9. (**a**) Antenna gain and (**b**) antenna efficiency vs. frequency.

1.1λ, respectively—were tested. As the DRA height increases, higher-order modes are more efficiently excited, resulting in higher gain, as discussed earlier.

Experimental validation

This section presents the measured results and compares them in detail with the simulated ones. The prototype antenna was fabricated and tested both without a reflector and for the higher-order mode HEM_{115}. A good agreement was observed between simulation and experimental results for the reflection coefficient, gain, and axial ratio, as discussed below. A prototype of the DRA was fabricated as shown in Fig. 14: (a) the DRA, (b) top view of the ground plane, and (c) the feed line. The resonator was produced using a 3-D printer with PLA material, and two PLA alignment arms of negligible thickness were attached to the DRA to ensure proper positioning with the elliptical slots, without affecting the antenna's electromagnetic performance.

Figure 15 compares the simulated and measured antenna axial ratio versus frequency. For the measurement, the same LP horn probe was used, and both amplitude and phase data were recorded for the co-polar and cross-polar linear components. The data were then postprocessed to obtain RHCP and LHCP at the desired theta angle ($\theta = 0°$). The results show similar ARBWs behaviour below 3 dB, indicating good circular polarization performance, with the AR range spanning 4.7–6.4 GHz, covering the WPT frequency. The measured results validate the design and confirm its ability to achieve circular polarization over the intended frequency band.

We also measured the reflection coefficient as a function of frequency. The experimental $|S_{11}|$ and the CP bandwidth are in close agreement, as shown in Fig. 16. The figure presents the simulated and measured – 10-dB impedance bandwidths, demonstrating a broadband performance of 64% from 3.3 GHz to 6.4 GHz, which represents a significant improvement over the previous design. Figure 17 shows the measured realized gain compared with the simulation results, demonstrating a good agreement across the operating band, with a peak gain of about 9.5 dBic. Minor discrepancies between simulated and measured gains can be attributed to surface finish or fabrication inaccuracies. A slight gain reduction is observed at higher frequencies (above 6.5 GHz) in both simulated and measured data, due to the mismatch losses. However, as shown in Fig. 8, adding the reflector significantly improved the peak gain, increasing it to 11.1 dBic.

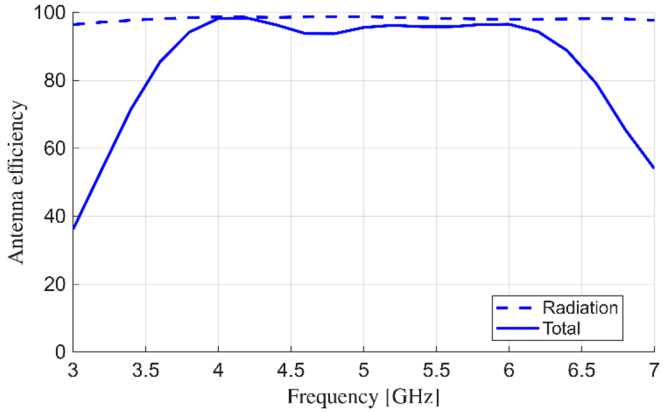


Fig. 9. (continued)

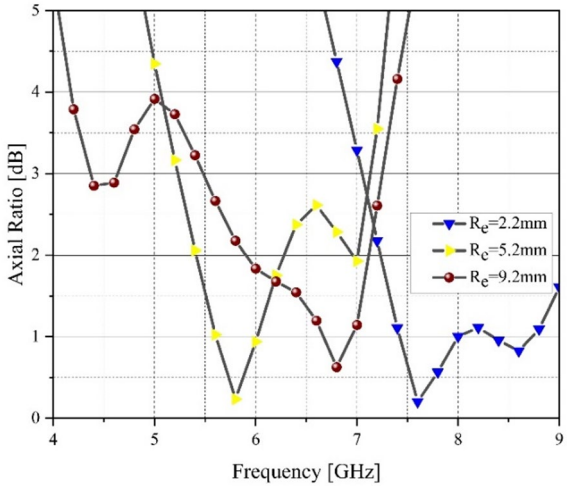


Fig. 10. Impact of changing elliptical radius.

The measured radiation patterns were obtained in an anechoic chamber using a standard far-field measurement setup. Patterns were measured in the E-plane and H-plane across the entire operating band. As shown in Fig. 18a, the radiation pattern at the centre frequency (5.8 GHz) exhibits the peak gain at $\theta = 0°$. Measurements at other frequencies, 4.5 GHz and 6 GHz, are shown in Fig. 18b, c, respectively, confirming the stability of the CP radiation patterns and the accuracy of the design with a cross-polar discrimination (XPD) greater than 20 dB.

Calculating the link budget for this antenna with an effective isotropic radiated power (EIRP) of 50 dBm (below the maximum allowed EIRP of 53 dBm at 5.8 GHz[38]) produces the response shown in Fig. 19. A received power of approximately 13 dBm is expected at a distance of 100 cm, and 0 dBm at 500 cm, which is above the typical sensitivity of commercial energy-harvesting circuits[39]. With this input RF power range, the harvesting circuit is expected to operate with a conversion efficiency above 50%, enabling rapid charging of the sensor's battery within a few minutes.

Finally, Table 1 compares the proposed CP DRA with earlier works, noting that coloured cells highlight drawbacks of existing antennas. The proposed CP DRA demonstrates outstanding performance, particularly in terms of:

- Highest gain among all compared designs (11.1 dBi), outperforming previously reported antennas. While similar gains are often achieved using multiple elements, our design achieves this with a single element.
- Wide impedance bandwidth, with an impedance BW of 64%, which is among the widest in the comparison.
- Wide axial ratio bandwidth of 31%, significantly larger than most reported CP antennas.
- Circular polarization is achieved with a simple single-feed elliptical slot, avoiding complex multi-feed or multi-port networks.
- Low-cost and simple fabrication, as the radiator is 3D-printed in a single piece using PLA and a standard 3-D printer, making it suitable for rapid, low-cost production.
- The proposed CP DRA has some limitations, such as:

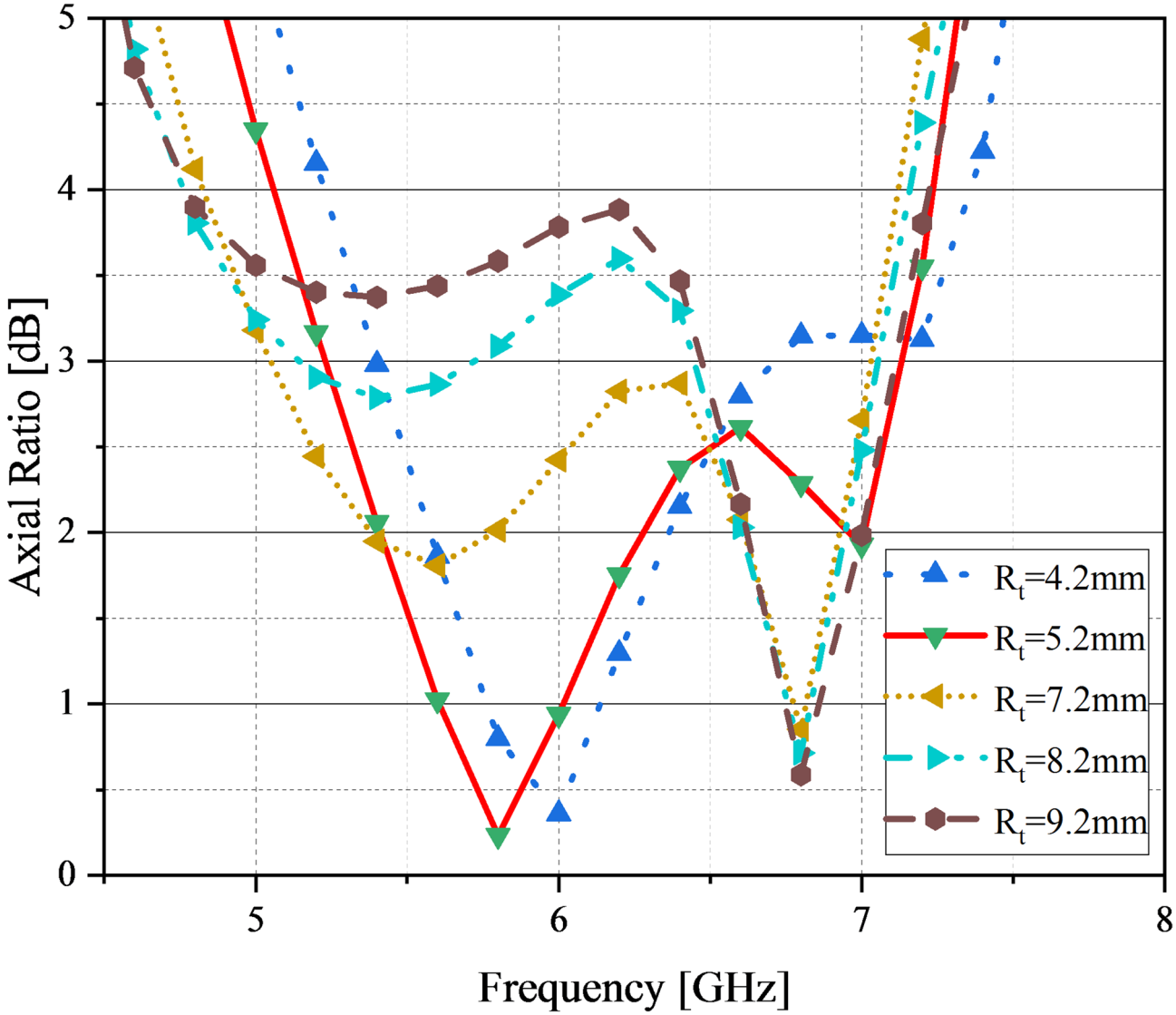


Fig. 11. Axial Ratio position with changing torus radius.

- Operates at a single centre frequency (5.8 GHz), whereas some of the mentioned works cover multiple separated bands.
- Larger assembly and integration complexity compared to fully planar or printed antennas.
- 3D-printing requires care, as the printing parameters (such as printing temperature and layer thickness) affect the dielectric properties.

Conclusion

This paper presents a low-cost circularly polarized DRA with wideband and high gain for WPT applications. The proposed design features a novel cupped-cone radiator with two-cylinder holes drilled into the DRA to enhance bandwidth and control the AR position. A novel feeding structure, based on a modified microstrip with loading stubs as well as elliptical cross and circular slots, is employed to increase the AR bandwidth. These enhancements achieve a good balance between gain (above 11 dBic), bandwidth (64%), and CP performance (AR-BW of 31%), along with clean radiation patterns featuring high SLL, XPD, and FBR, making the antenna suitable for microwave WPT applications. The design is particularly relevant where low cost and ease of fabrication are essential, such as battery-free wireless sensors for environmental monitoring and smart homes/cities. The DRA bandwidth fully covers the WPT ISM band at 5.8 GHz. A 3D-printed prototype was fabricated, and the experimental results show excellent agreement with simulations.

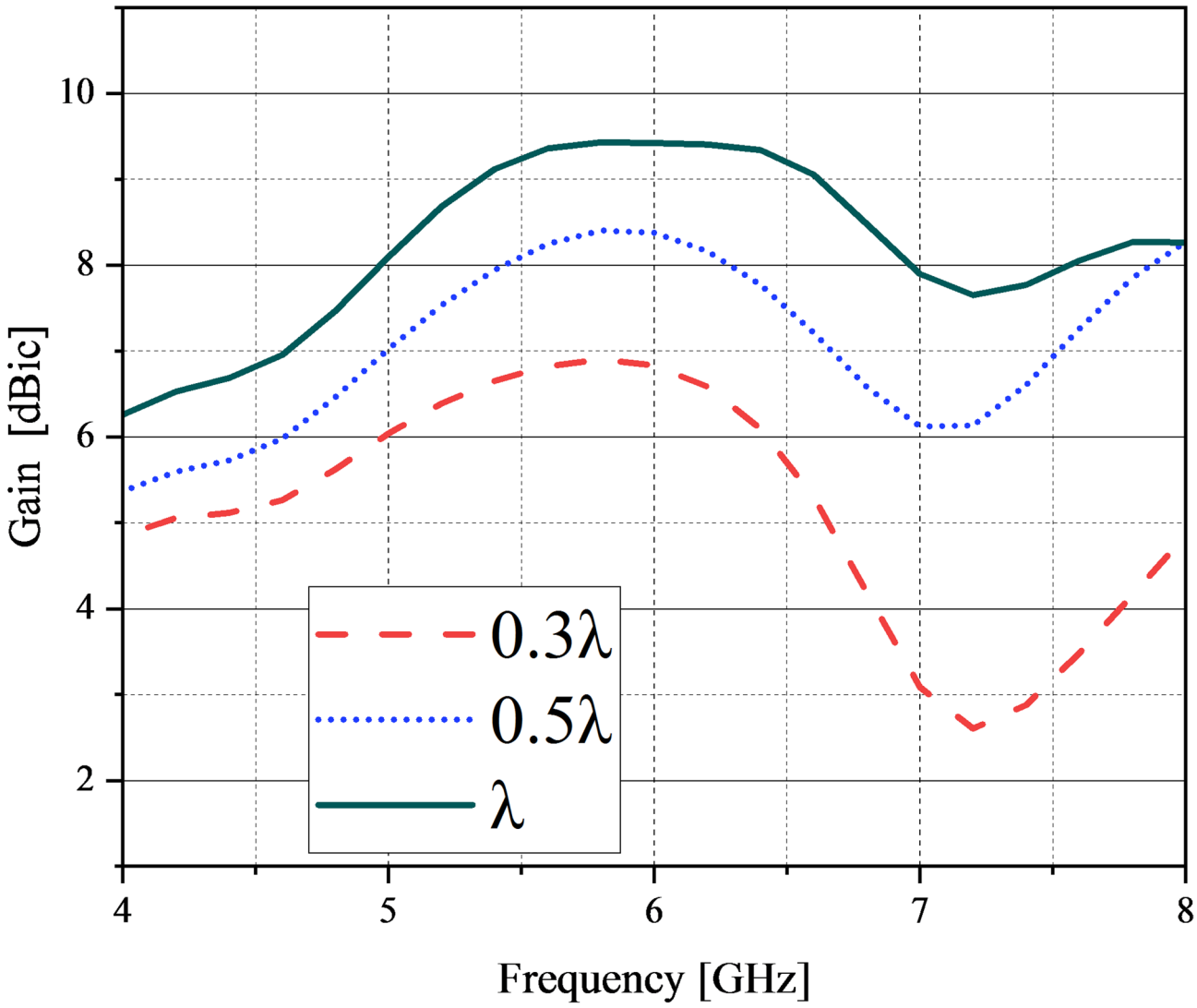


Fig. 12. Gain vs. frequency with increasing DRA height.

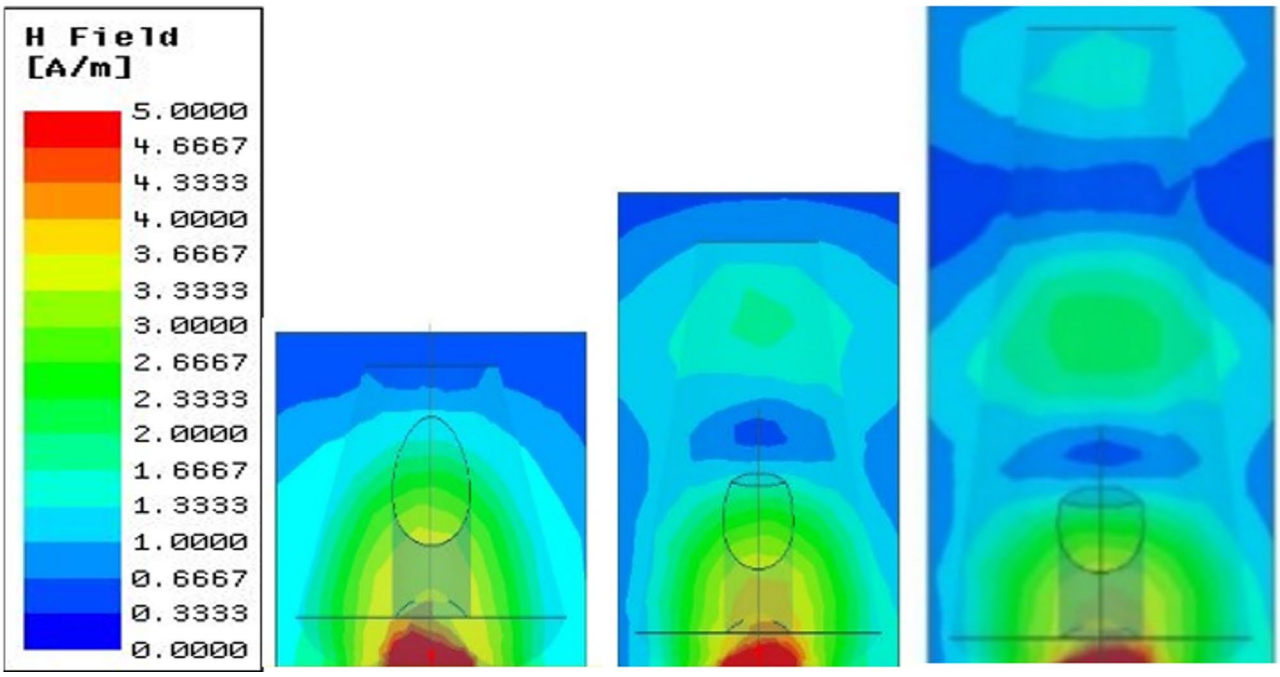


Fig. 13. H field distribution at 5.8 GHz for different DRA heights.



(a) (b) (c)

Fig. 14. Fabricated prototype (**a**) 3D photograph of the DRA, (**b**) elliptical slot aperture, and (**c**) microstrip feed line.

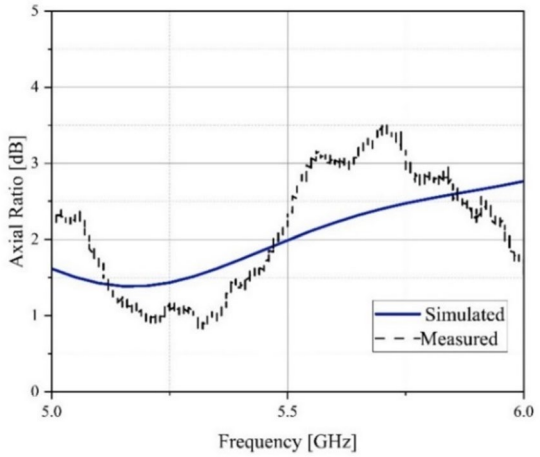


Fig. 15. Simulations and measurements of axial ratio vs. frequency.

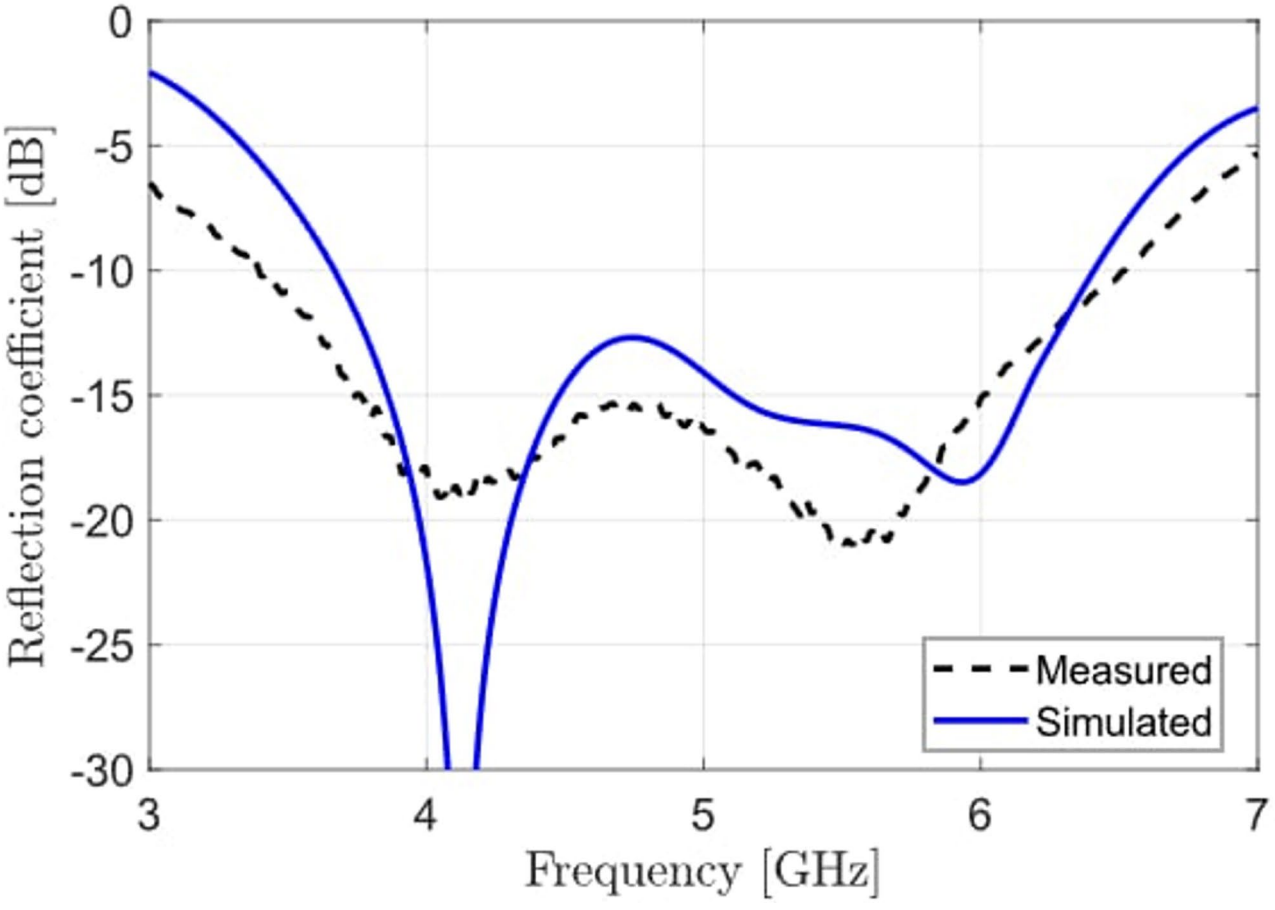


Fig. 16. Measured and simulated reflection coefficients.

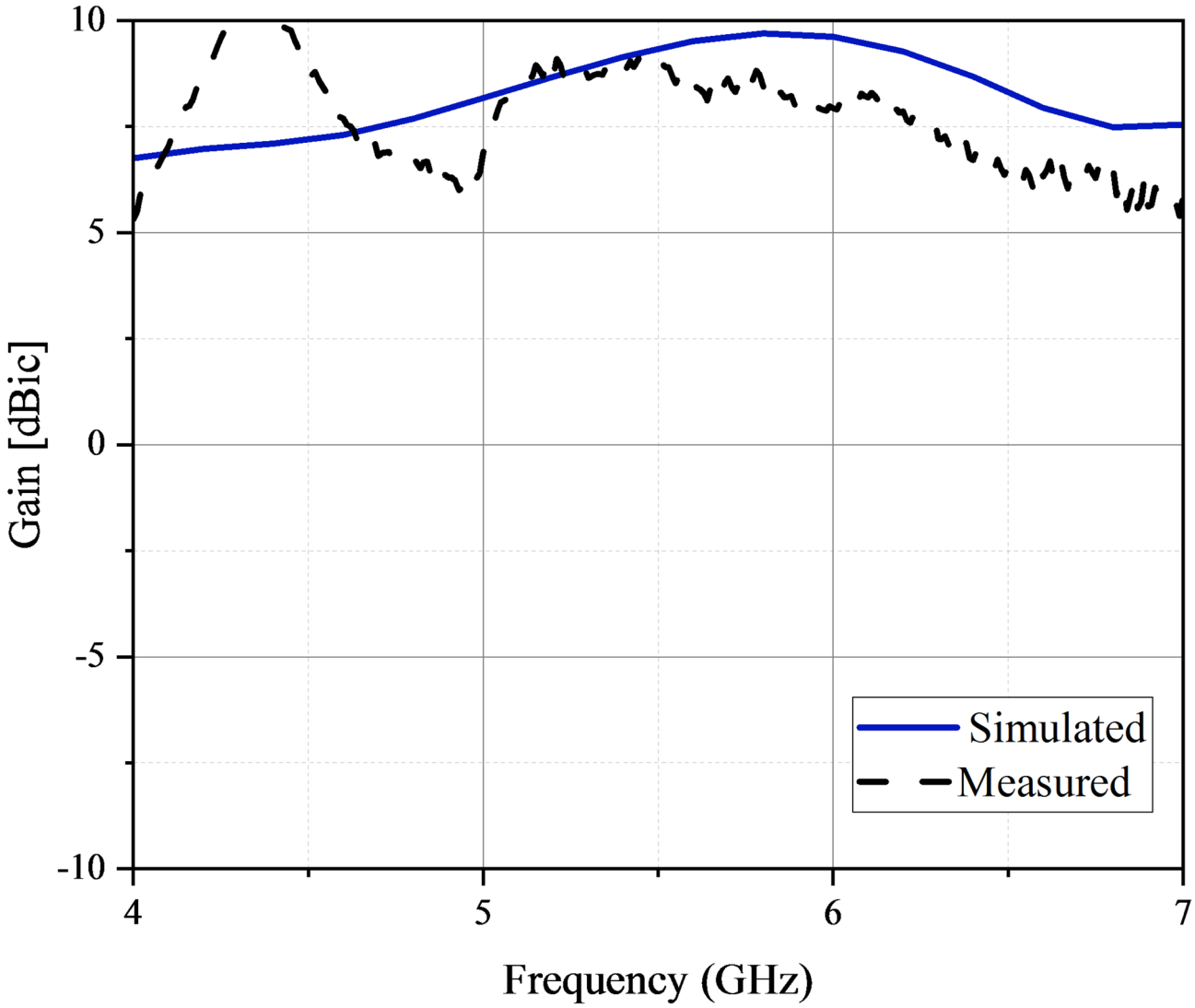


Fig. 17. Measured and simulated gain vs. frequency.

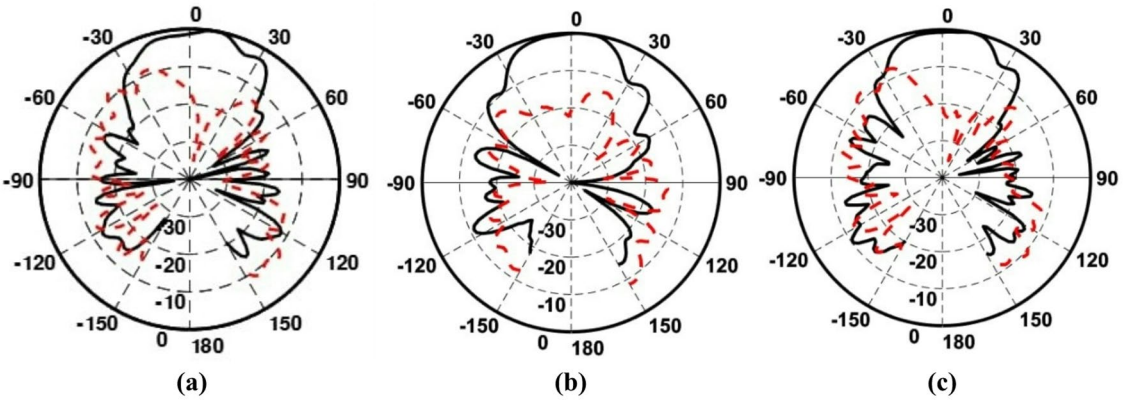


Fig. 18. Normalized measured radiation patterns at (**a**) 5.8 GHz, (**b**) 4.5 GHz, and (**c**) 6 GHz.

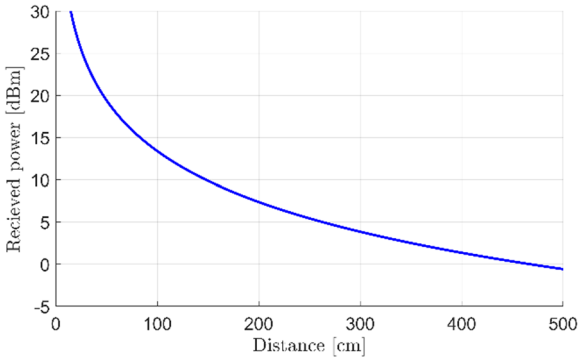


Fig. 19. Expected link budget for WPT experiment.

Ref.	Radiator	Material	Operating frequency	Feeding Tech.	Polarization	Gain (dBi)	– 10 dB IBW%	3 dB AR-BW%
[8]/2025	Rectangular	PLA $\varepsilon_r = 2.7$	26.2 GHz	Cross microstrip line	LP	6.8	22.5	–
[9]/2019	Rectangle with a metal strip	$\varepsilon_r = 10$	3.5–4.5 GHz	E-shape metal strip	CP	6.5	23	22
[37]/2022	Hypered cylinder	ECCOSTOCK HIK $\varepsilon_r = 20$	10 GHz	Two-point feeding	CP	7.7	34	23
[40]/2021	YDRA	Alumina $\varepsilon_r = 10$	2.4–3.5 GHz	Vertical strip	CP	4.8	53.5	10
[41]/2021	Rectangular	HIKK10 $\varepsilon_r = 10$	5.58–5.99 GHz	Slot-loaded ground	CP	6.41	7	4
[42]/2022	Sheets and strips	Ceramic $\varepsilon_r = 45$ for sheets, $\varepsilon_r = 69$ for strips	10 GHz	Rectangle slots	CP	6.6	17.5	12
[43]/2023	Rectenna	Metal	4.1–6.6 GHz	Five-port	CP	7.8	46.3	10
[44]/2024	Groove	Metal	5.8 GHz	Microstrip Line	LP	9.4	12	–
[45]/2025	L-shape	PLA + MG metal spray	10 GHz	Waveguide	CP	7.5	7.4	7
[46]/2025	H-shape	Metal	5.8 GHz	Microstrip line	CP	3.3	6	8.7
[47]/2025	Shared aperture	Metal + DR $\varepsilon_r = 10$	20–30 GHz	Dual port	CP	6.4	17.5	7.5
[48]/2025	4-elements	Metal	6 GHz	MIMO (4-port)	CP	7.9	64.5	54.5
[49]/2025	2-elements	Alumina ceramic	28 GHz	Metasurface lens	CP	11	4	2.3
Prop.	Cupped-cone shape	PLA $\varepsilon_r = 2.7$	5.8 GHz	Elliptical slot	CP	11.1	64	31

Table 1. Comparison between the proposed design and state-of-the-art solutions. Coloured cells highlight drawbacks of existing antennas.

Data availability

No dataset is used in this study. The full experimental and simulated data are available in the manuscript.

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Declarations

Competing interests
The authors declare no competing interests.

Additional information
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